MNC UNO R1

An electronics development board from MENOCLONE Motorsports.



LEARN THE CIRCUIT. BUILD THE IDEA.

The MNC UNO R1 is a development board concept created to make electronics more approachable. It is intended for hands-on learning with circuits, sensors, programming, and small hardware projects.

Inspired by the familiar Arduino Uno style of development, the board is part of MENOCLONE Motorsports' effort to connect practical engineering skills with original product design.

WHY WE ARE BUILDING IT

MENOCLONE Motorsports began with curiosity about how electronic systems work. The UNO R1 extends that interest into a board that learners can use to experiment, make mistakes, and understand the relationship between code and physical components.

Development note: the board is a proposed design. Final components, electrical limits, and compatibility require validation.

What a development board does

A small platform for exploring the connection between software and electronics.

A microcontroller runs a program that reads inputs and controls outputs. A development board makes those connections easier to reach, so a learner can work with switches, LEDs, sensors, and other modules without designing an entire circuit from the beginning.

CORE FUNCTIONS TO EXPLORE

Digital input and output
Read a button, switch an LED, or control a compatible external circuit.

Analog measurement
Observe a changing sensor voltage through supported analog inputs.

Timing and control
Use code to coordinate blinking, counting, and timed responses.

Serial communication
Exchange messages between the board and a computer.

External modules
Connect suitable sensors and modules while respecting their voltage and current requirements.

Exact pin assignments and electrical ratings will be published after the hardware design is verified.

From first blink to working prototype

A progression designed for makers, students, and independent developers.

01 / FIRST CIRCUIT

Connect an LED through an appropriate resistor and write a program to switch it on and off.

02 / READ AN INPUT

Use a pushbutton and observe how the program responds to a change in its input state.

03 / MEASURE A SENSOR

Read a compatible sensor, convert its output into a useful value, and display the result.

04 / COMBINE HARDWARE AND CODE

Build a small monitoring or control project using inputs, outputs, and a clear program structure.

Good practice: Check supply voltage, component polarity, and current limits before powering a circuit. Disconnect power when changing wiring.

04 / DESIGN AND TESTING

Building the MNC UNO R1

From schematic review to a board that can be used with confidence.

Before the board can be released, its schematic, PCB layout, component choices, and documentation must be checked together. A visually complete design is not the same as a tested electronic product.

DEVELOPMENT CHECKPOINTS

01 Schematic review
Confirm the controller, power supply, clock, programming interface, connectors, and protection circuits.

02 PCB inspection
Check footprints, trace routing, grounding, and assembly clearances.

03 Power-on testing
Measure the power rails and confirm expected behavior before connecting external modules.

04 Firmware tests
Upload basic programs and verify input, output, and communication functions.

05 Documentation
Publish validated specifications, pin information, setup steps, and known limitations.

The final bill of materials, board revision, and verified specifications will be defined during prototype testing.

05 / FUTURE DIRECTION

A board for curious minds

Independent electronics development is part of the larger MENOCLONE Motorsports vision.

The MNC UNO R1 is intended to be more than a branded circuit board. It represents the habit of learning through building: understanding a component, testing an idea, and refining the result.

PART OF A CONNECTED WORKSHOP

MNC UNO R1 supports foundational electronics learning. **MNC DIAG-01** explores automotive diagnostic communication. Both projects reflect an interest in hardware, software, and practical problem solving.

As the MENOCLONE Motorsports Auto Club grows, these projects can help introduce more people to electronics and the technology inside modern vehicles.

ORIGINALITY HAS ORIGINS.

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